ELEMENTARY BOUNDS FOR LAPLACE APPROXIMATION OF PROBABILITY MEASURES

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Abstract. We present short proofs of non-asymptotic bounds for Gaussian (Laplace) approximation to a smooth probability measure on a Euclidean space, with errors quantified in total variation, relative entropy, Wasserstein distance, and accuracy of normalising constant estimation. The main bounds are expressed in terms of the Poincaré and log-Sobolev constants of the target measure, with estimates based on uniform quadratic convexity of the target potential then deductible as simple sufficient conditions. Granted suitable smoothness assumptions and Gaussian moment estimates, one recovers the familiar $d/\sqrt{n}$ scaling for the errors. We also discuss extensions based on local functional inequalities, effective dimension, and sketch a small statistical example.

The results presented here are largely expository reformulations of existing results and arguments in the literature on functional inequalities and quantitative Laplace approximation. Our emphasis is on short, elementary proofs using functional inequalities, with explicit constants.

1. Functional inequalities and the score error

Let $U : \mathbb{R}^d \to \mathbb{R}$ be smooth, translated and whitened so that

$$U(0) = 0, \qquad \nabla U(0) = 0, \qquad \nabla^2 U(0) = I_d,$$

and write

$$\mu(dx) = Z^{-1} e^{-U(x)}\, dx, \qquad \gamma = \mathcal{N}(0, I_d), \qquad r(x) = U(x) - \tfrac{1}{2}|x|^2.$$

We take as given that $Z < \infty$; write also the normalising constant for the Gaussian approximation as $Z_{\mathrm{Lap}} = (2\pi)^{d/2}$.

For a probability measure ν, let $C_{\mathrm{PI}}(\nu)$ and $C_{\mathrm{LSI}}(\nu)$ be the *optimal* constants (possibly infinite) in the comparisons (for all e.g. locally Lipschitz functions h)

$$\mathrm{Var}_\nu(h) \le C_{\mathrm{PI}}(\nu)\, \mathbb{E}_\nu |\nabla h|^2, \tag{1.1}$$

$$\mathrm{Ent}_\nu(h^2) \le 2 C_{\mathrm{LSI}}(\nu)\, \mathbb{E}_\nu |\nabla h|^2, \tag{1.2}$$

where $\mathrm{Ent}_\nu(g) = \mathbb{E}_\nu[g \log(g/\mathbb{E}_\nu g)]$. Our convention is such that $C_{\mathrm{PI}}(\gamma) = C_{\mathrm{LSI}}(\gamma) = 1$ and smaller constants map onto better-concentrated measures. If $B_R = \{|x| \le R\}$ and $\nu(B_R) > 0$, we also write

$$\nu_R = \nu(\,\cdot \mid B_R), \qquad C_{\mathrm{PI}}(\nu; R) = C_{\mathrm{PI}}(\nu_R), \qquad C_{\mathrm{LSI}}(\nu; R) = C_{\mathrm{LSI}}(\nu_R)$$

for the constants associated to the constrained measures. Finally, set

$$M_3(U) = \sup_x \|\nabla^3 U(x)\|_{\mathrm{op}}, \qquad M_4(U) = \sup_x \|\nabla^4 U(x)\|_{\mathrm{op}},$$

where the multilinear operator norm takes a supremum over unit vectors in every argument; $M_3(U; R)$ and $M_4(U; R)$ denote the same suprema restricted to B_R.

A recurring quantity of interest will be the *Gaussian-averaged score error*

$$\mathcal{S}(U) = \Big(\mathbb{E}_\gamma |\nabla r(G)|^2\Big)^{1/2} = \Big(\mathbb{E}_\gamma |\nabla U(G) - G|^2\Big)^{1/2}.$$

Following its name, this is an average against the standard Gaussian, and is hence reasonably estimable. Moreover, the use of derivatives obviates the need to explicitly estimate the normalising constant Z; this will prove useful in subsequent developments.

Lemma 1. *(Poincaré–Hellinger comparison) If μ, ν have positive smooth densities, then*

$$\mathrm{TV}(\mu, \nu) \le \frac{\sqrt{C_{\mathrm{PI}}(\mu)}}{2} \left(\mathbb{E}_\nu \left| \nabla \log \frac{d\nu}{d\mu} \right|^2 \right)^{1/2}. \tag{1.3}$$

This inequality is a special case of [3, Theorem 3.1(a)].

Proof. Set $h = \sqrt{d\nu/d\mu}$ and $a = \mathbb{E}_\mu h$. The Cauchy–Schwarz inequality gives that

$$\mathrm{TV}(\mu,\nu) = \tfrac{1}{2}\mathbb{E}_\mu|1-h^2| \le \tfrac{1}{2}\sqrt{\mathbb{E}_\mu(1-h)^2\,\mathbb{E}_\mu(1+h)^2} = \sqrt{1-a^2} = \sqrt{\mathrm{Var}_\mu(h)},$$

where the last quantity relates to the Hellinger distance between μ and ν.Now apply (1.1) and use $\mathbb{E}_\mu|\nabla h|^2 = \frac{1}{4}\mathbb{E}_\nu|\nabla \log(d\nu/d\mu)|^2$. □

Lemma 2. *(Log-Sobolev comparison) For the same pair of measures,*

$$\mathrm{KL}(\nu\|\mu) \le \frac{C_{\mathrm{LSI}}(\mu)}{2}\mathbb{E}_\nu\left|\nabla \log\frac{d\nu}{d\mu}\right|^2. \tag{1.4}$$

If $C_{\mathrm{LSI}}(\mu) < \infty$, one also has

$$W_2(\nu,\mu) \le C_{\mathrm{LSI}}(\mu)\left(\mathbb{E}_\nu\left|\nabla \log\frac{d\nu}{d\mu}\right|^2\right)^{1/2}. \tag{1.5}$$

The relative entropy inequality is simply the logarithmic Sobolev inequality applied to the square-root likelihood ratio. The Wasserstein bound additionally uses the classical implication from logarithmic Sobolev to quadratic transport inequalities; see Guillin et al. [3, Proposition 2.11]. *For applications of these inequalities to quantitative Laplace approximation, see Kasprzak et al.* [4, Section 5.1].

Proof. Apply (1.2) to $h = \sqrt{d\nu/d\mu}$: $\mathrm{Ent}_\mu(h^2) = \mathrm{KL}(\nu\|\mu)$ and $\mathbb{E}_\mu|\nabla h|^2$ is the same quantity as in Lemma 1. For (1.5), combine this with the standard LSI-to-transport inequality $W_2^2(\nu,\mu) \le 2C_{\mathrm{LSI}}(\mu)\,\mathrm{KL}(\nu\|\mu)$. □

In particular, setting $\nu = \gamma$, we obtain the abstract bounds.

$$\mathrm{TV}(\mu,\gamma) \le \tfrac{1}{2}\sqrt{C_{\mathrm{PI}}(\mu)}\,\mathcal{S}(U), \qquad \mathrm{KL}(\gamma\|\mu) \le \tfrac{1}{2}C_{\mathrm{LSI}}(\mu)\,\mathcal{S}(U)^2. \tag{1.6}$$

2. Error Bounds for Measure approximation

Theorem 3. *Suppose $M_3(U) < \infty$ and the constants below are finite. Then*

$$\mathrm{TV}(\mu,\gamma) \le \frac{M_3(U)}{4}\sqrt{C_{\mathrm{PI}}(\mu)\,d(d+2)}, \tag{2.1}$$

$$\mathrm{KL}(\gamma\|\mu) \le \frac{M_3(U)^2}{8}C_{\mathrm{LSI}}(\mu)\,d(d+2), \tag{2.2}$$

$$\mathrm{KL}(\mu\|\gamma) \le \frac{M_3(U)^2}{8}\mathbb{E}_\mu|X|^4, \tag{2.3}$$

$$W_2(\mu,\gamma) \le \frac{M_3(U)}{2}C_{\mathrm{LSI}}(\mu)\sqrt{d(d+2)}. \tag{2.4}$$

The bound on W_2 also bounds $|\mathbb{E}_\mu X|$.

The total variation estimate is a particularly simple instance of the Fisher information approach to Laplace approximation developed by Kasprzak et al. [4] *and Katsevich* [5]*; see also Katsevich and Spokoiny* [7, Lemma 2.8]. *The remaining bounds follow from the same score estimate by standard functional inequalities.*

Proof. Since $\nabla^2 r(0) = 0$, Taylor's theorem gives

$$|\nabla r(x)| \le \int_0^1 \|\nabla^2 U(tx) - I_d\|\,|x|\,dt \le \tfrac{1}{2}M_3(U)|x|^2.$$

Thus $\mathcal{S}(U)^2 \le \frac{1}{4}M_3(U)^2\mathbb{E}|G|^4$, and $\mathbb{E}|G|^4 = d(d+2)$. Equations (1.6) and (1.5) yield the first, second and fourth bounds. For the reverse KL bound, apply Gaussian LSI to bound

$$\mathrm{KL}(\mu\|\gamma) \le \tfrac{1}{2}\mathbb{E}_\mu|\nabla r(X)|^2 \le \tfrac{1}{8}M_3(U)^2\mathbb{E}_\mu|X|^4.$$

□

Remark 4. If $\nabla^2 U \succeq \alpha I_d$ for some $\alpha > 0$, then the Bakry–Émery criterion gives $C_{\mathrm{PI}}(\mu) \le C_{\mathrm{LSI}}(\mu) \le 1/\alpha$. Integration by parts also gives the elementary moment estimate $\mathbb{E}_\mu|X|^4 \le d(d+2)/\alpha^2$.Substituting into Theorem 3 recovers the bounds $M_3(U)\sqrt{d(d+2)}/(4\sqrt{\alpha})$ for TV, $M_3(U)^2 d(d+2)/(8\alpha)$ and $M_3(U)^2 d(d+2)/(8\alpha^2)$ for the two KL directions, and $M_3(U)\sqrt{d(d+2)}/(2\alpha)$ for W_2.

As a consequence of the W_2 estimate, we can also control the accuracy of the approximation to the covariance matrix of μ.

Corollary 5. *(Covariance approximation). Under the assumptions of Theorem 1, set*

$$\varepsilon := \frac{M_3(U)}{2} C_{\mathrm{LSI}}(\mu)\sqrt{d(d+2)}.$$

Then

$$\|\mathrm{Cov}_\mu(X) - I\|_{\mathrm{op}} \le 2\varepsilon + \varepsilon^2.$$

In particular, if $\varepsilon < 1$, then

$$(1-\varepsilon)^2 I \preceq \mathrm{Cov}_\mu(X) \preceq (1+\varepsilon)^2 I.$$

Proof. (Sketch) For any unit vector v, the triangle inequality in L^2, applied to centred random variables under a coupling of $X \sim \mu$ and $G \sim \gamma$, gives $\left|\sqrt{\mathrm{Var}_\mu(v^\top X)} - 1\right| \le W_2(\mu,\gamma) \le \varepsilon$. The claims then follow by squaring and taking the supremum over unit vectors. □

Remark 6. The direct cubic Taylor bound $|r(G)| \le M_3(U)|G|^3/6$ suggests the dimension scaling $d^{3/2}$ for the error. Taking one derivative and applying the Poincaré inequality improves this to $\sqrt{\mathbb{E}|G|^4} = \sqrt{d(d+2)}$, whence the improved dimension scaling d emerges.

3. Error Bounds for Normalising Constant Approximation

The identity $Z/Z_{\mathrm{Lap}} = \mathbb{E}_\gamma e^{-r(G)}$ gives

$$\log \frac{Z}{Z_{\mathrm{Lap}}} = \mathrm{KL}(\gamma\|\mu) - \mathbb{E}_\gamma r(G). \tag{3.1}$$

Proposition 7. *(Normalising constant error) If $C_{\mathrm{LSI}}(\mu) < \infty$ and $M_3(U), M_4(U) < \infty$, then*

$$\left|\log \frac{Z}{Z_{\mathrm{Lap}}}\right| \le d(d+2)\left(\frac{M_4(U)}{24} + \frac{C_{\mathrm{LSI}}(\mu) M_3(U)^2}{8}\right). \tag{3.2}$$

The cancellation of the cubic term is a familiar feature of higher-order Laplace expansions. Related refinements exploiting Gaussian symmetry appear in Spokoiny [8, Theorem 2.2]. *The present estimate follows directly from the entropy identity above and does not require a higher-order expansion of the normalising constant.*

Proof. Write

$$r(x) = \tfrac{1}{6}\nabla^3 U(0)[x,x,x] + R_4(x), \qquad |R_4(x)| \le \tfrac{1}{24} M_4(U)|x|^4.$$

The cubic term is odd, so its Gaussian expectation is zero. Thus $|\mathbb{E}_\gamma r| \le M_4(U) d(d+2)/24$. Combine (3.1) and (2.2). □

The vanishing of the Gaussian cubic moment explains why the log-normalising-constant bound is second-order in the small-derivative parameter, even though the TV error is first-order.

4. Scaling with Sample Size in Statistical Models

Let

$$\pi_n(d\theta) = Z_n^{-1} e^{-n f(\theta)}\, d\theta, \qquad \hat\theta = \arg\min f, \qquad H = \nabla^2 f(\hat\theta) \succ 0,$$

and $\gamma_n = \mathcal{N}(\hat\theta, (nH)^{-1})$. Define

$$\delta_k(f;H) = \sup_\theta \sup_{|u_1|,\dots,|u_k|\le 1} \left|\nabla^k f(\theta)[H^{-1/2}u_1, \dots, H^{-1/2}u_k]\right|, \quad k = 3,4.$$

Let μ_n be the law of $X = \sqrt{n}\, H^{1/2}(\Theta - \hat\theta)$ under $\Theta \sim \pi_n$, whose potential U_n satisfies

$$M_3(U_n) = \frac{\delta_3(f;H)}{\sqrt{n}}, \qquad M_4(U_n) = \frac{\delta_4(f;H)}{n}.$$

Recall $C_{\mathrm{PI}}(\mu_n)$ and $C_{\mathrm{LSI}}(\mu_n)$ as the geometric constants associated to the *whitened posterior*. Theorem 3 then becomes

$$\mathrm{TV}(\pi_n, \gamma_n) \le \frac{\delta_3(f;H)}{4}\sqrt{\frac{C_{\mathrm{PI}}(\mu_n)\, d(d+2)}{n}}, \tag{4.1}$$

$$\mathrm{KL}(\gamma_n\|\pi_n) \le \frac{\delta_3(f;H)^2}{8}\frac{C_{\mathrm{LSI}}(\mu_n)d(d+2)}{n}, \tag{4.2}$$

$$\mathrm{KL}(\pi_n\|\gamma_n) \le \frac{\delta_3(f;H)^2}{8}\frac{\mathbb{E}_{\mu_n}|X|^4}{n}. \tag{4.3}$$

For the metric $|\theta-\theta'|_H = |H^{1/2}(\theta-\theta')|$,

$$W_{2,H}(\pi_n,\gamma_n) \le \frac{\delta_3(f;H)}{2}\frac{C_{\mathrm{LSI}}(\mu_n)\sqrt{d(d+2)}}{n}. \tag{4.4}$$

The same right-hand side bounds $|H^{1/2}(\mathbb{E}_{\pi_n}\Theta-\hat\theta)|$. Finally, with $Z_{n,\mathrm{Lap}} = e^{-nf(\hat\theta)}(2\pi/n)^{d/2}\det(H)^{-1/2}$,

$$\left|\log\frac{Z_n}{Z_{n,\mathrm{Lap}}}\right| \le \frac{d(d+2)}{n}\left(\frac{\delta_4(f;H)}{24}+\frac{C_{\mathrm{LSI}}(\mu_n)\delta_3(f;H)^2}{8}\right). \tag{4.5}$$

If $\nabla^2 f \succeq \alpha H$ globally, then $C_{\mathrm{PI}}(\mu_n), C_{\mathrm{LSI}}(\mu_n) \le 1/\alpha$ and $\mathbb{E}_{\mu_n}|X|^4 \le d(d+2)/\alpha^2$, recovering fully explicit $d/\sqrt{n}$ and d^2/n bounds as an immediate corollary.

5. Local comparison and effective dimension

Often only a region near the mode has well-controlled curvature. This motivates a modified analysis based around use of the constrained Poincaré constant $C_{\mathrm{PI}}(\mu;R)$.

Proposition 8. *(Local comparison) Let $B_R=\{|x|\le R\}$, suppose $\mu(B_R),\gamma(B_R)>0$, and assume $M_3(U;R)<\infty$ and $C_{\mathrm{PI}}(\mu;R)<\infty$. Then*

$$\mathrm{TV}(\mu,\gamma) \le \mu(B_R^c)+\gamma(B_R^c)+\frac{M_3(U;R)}{4}\sqrt{C_{\mathrm{PI}}(\mu;R)}\left(\mathbb{E}_\gamma[|G|^4 \mid |G|\le R]\right)^{1/2}. \tag{5.1}$$

If $\gamma(B_R)\ge\frac12$, the last term is at most $\frac{M_3(U;R)}{2\sqrt{2}}\sqrt{C_{\mathrm{PI}}(\mu;R)d(d+2)}$.

This local-to-global decomposition follows Katsevich [5, Lemma 2.11]*; see also Katsevich and Spokoiny* [7, Lemmas 2.4 and 2.8]*. Under global convexity, the required tail estimates can be obtained from local curvature and linear growth away from the mode, as in Katsevich* [5, Lemma 2.10 and Theorem 2.2]*.*

Proof. Apply Lemma 1 to the conditional laws μ_R,γ_R. Their log-density ratio differs from the unconditioned one only by a constant, which then vanishes upon differentiation. Use $|\nabla r(x)|\le M_3(U;R)|x|^2/2$ inside the ball, and then remove conditioning at a cost at most $\mu(B_R^c)+\gamma(B_R^c)$. □

For example, $R=\sqrt{d}+\sqrt{2t}$ gives $\gamma(B_R^c)\le e^{-t}$. If $\nabla^2U\succeq\alpha I_d$ *only inside* B_R, then the constrained Bakry–Émery inequality on a convex domain still gives $C_{\mathrm{PI}}(\mu;R), C_{\mathrm{LSI}}(\mu;R)\le 1/\alpha$. The remaining issue is the target tail $\mu(B_R^c)$; global convexity and local curvature can be used to control it, as in Katsevich [5]. The conditional log-Sobolev constant also yields the local KL comparison $\mathrm{KL}(\gamma_R\|\mu_R)\le\frac12 C_{\mathrm{LSI}}(\mu;R)\mathbb{E}_{\gamma_R}|\nabla r|^2$, if desired.

A second refinement allows anisotropic geometry. For $D\succ 0$ define weighted constants by

$$\mathrm{Var}_\mu(h)\le C_{\mathrm{PI}}(\mu;D)\mathbb{E}_\mu|D^{-1}\nabla h|^2, \quad \mathrm{Ent}_\mu(h^2)\le 2C_{\mathrm{LSI}}(\mu;D)\mathbb{E}_\mu|D^{-1}\nabla h|^2.$$

Suppose $\gamma=\mathcal{N}(0,H^{-1})$, $\nabla U(0)=0$, $\nabla^2U(0)=H$, and

$$\left\|D^{-1}(\nabla^2U(x)-H)D^{-1}\right\|_{\mathrm{op}}\le\tau|Dx|.$$

Writing $S=DH^{-1}D$ and $p=\operatorname{tr}S$, the same Poincaré–Hellinger proof, now in the D-metric, gives

$$\mathrm{TV}(\mu,\gamma)\le\frac{\tau}{4}\sqrt{C_{\mathrm{PI}}(\mu;D)}\sqrt{p^2+2\operatorname{tr}(S^2)}. \tag{5.2}$$

Indeed, $|D^{-1}(\nabla U(x)-Hx)|\le\tau|Dx|^2/2$ and $\mathbb{E}_\gamma|DG|^4=p^2+2\operatorname{tr}(S^2)$. If $S\preceq I_d$, then $\operatorname{tr}(S^2)\le p$; if also $\nabla^2U\succeq\alpha D^2$, then $C_{\mathrm{PI}}(\mu;D)\le 1/\alpha$. The use of an auxiliary metric to obtain effective-dimensional Laplace bounds is developed systematically by Katsevich and Spokoiny [7, Theorem 2.5]. In particular, their proof uses the same Gaussian fourth-moment identity; the present formulation isolates the weighted Poincaré constant as a (slightly) more general sufficient condition.

6. A sketched statistical example

For a canonical generalised linear model with a quadratic prior,

$$f_n(\theta) = \frac{1}{n}\sum_{i=1}^{n}\{b(x_i^\top\theta) - y_i x_i^\top\theta\} + \frac{\lambda}{2n}|\theta|^2, \qquad \lambda \geq 0.$$

Write $H = \nabla^2 f_n(\hat{\theta}) \succ 0$, $z_i = H^{-1/2}x_i$, and $\rho = \max_i |z_i|$. For logistic regression, $b(t) = \log(1+e^t)$, and

$$|b'''(t)| \leq b''(t), \quad |b''''(t)| \leq b''(t), \quad |\{\log b''(t)\}'| \leq 1.$$

On $|\theta - \hat{\theta}|_H \leq s$, $b''(x_i^\top\theta) \leq e^{\rho s} b''(x_i^\top\hat{\theta})$. Since $\frac{1}{n}\sum_i b''(x_i^\top\hat{\theta}) z_i z_i^\top \preceq I_d$, Cauchy–Schwarz gives

$$\sup_{|\theta-\hat{\theta}|_H \leq s} \|\nabla^3 f_n(\theta)\|_H \leq \rho e^{\rho s}, \qquad \sup_{|\theta-\hat{\theta}|_H \leq s} \|\nabla^4 f_n(\theta)\|_H \leq \rho^2 e^{\rho s}.$$

Here $\|\cdot\|_H$ is the tensor norm after inserting $H^{-1/2}$ in every slot. This curvature comparison is an instance of the generalized self-concordance property of logistic regression; see Bach [1, Proposition 1, equation (6)].

These derivative estimates can be plugged into the local theorem once $C_{\mathrm{PI}}(\mu; R)$ and the target's tail are controlled. For Gaussian linear regression with a Gaussian prior, the potential is quadratic, so $M_3(U) = M_4(U) = 0$ and Laplace approximation is exact.

Remark 9. The earlier covariance estimate also applies in the original parameterisation. Writing

$$\varepsilon_n := \frac{\delta_3(f; H)}{2\sqrt{n}} C_{\mathrm{LSI}}(\mu_n)\sqrt{d(d+2)},$$

we obtain

$$\left\| nH^{1/2}\,\mathrm{Cov}_{\pi_n}(\Theta)H^{1/2} - I \right\|_{\mathrm{op}} \leq 2\varepsilon_n + \varepsilon_n^2.$$

As such, if the standardised smoothness and functional inequality constants remain bounded with n, then the relative covariance error is controlled at the same $d/\sqrt{n}$ scale as the total variation error.

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